

EEE554 – Probability and Random Processes

Module 1: Introduction to Probability and Set Theory

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Probability versus Statistics

A common question is:

What is the difference between probability and statistics?

Although probability and statistics are closely related, they address different problems involving uncertainty.

- Probability: Starts with a mathematical model of uncertainty and analyzes the likelihood of various outcomes.
- Statistics: Starts with observed data and uses it to infer the underlying probabilistic model or its unknown parameters.

In short,

Probability: Model $\longrightarrow$ Data

whereas

Statistics: Data $\longrightarrow$ Model.

Why Study Probability?

Probability theory provides the mathematical language for describing uncertainty. It forms the foundation of many areas in electrical and computer engineering.

Representative applications:

- Communications
- Information Theory
- Computer Networks
- Quantum Information Theory
- Quantum Mechanics

- Statistical Signal Processing
- Control Systems
- Electromagnetics and Optics
- Machine Learning and Artificial Intelligence

Throughout this course we will repeatedly see that probability provides models for uncertain phenomena, while statistics develops methods for learning those models from observations.

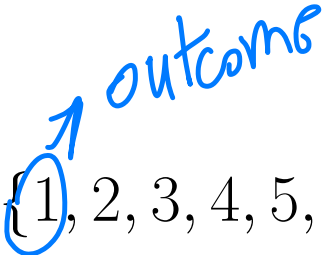
Review of Set Theory

Probability theory is built upon elementary set theory.

Why Set Theory?

Consider the experiment of rolling a fair die.

The set of all possible outcomes is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$


In probability, we are interested in asking questions about the outcome.

For example:

Is the outcome even? $A = \{2, 4, 6\}$.

$\omega \in A?$

Is the outcome odd? $B = \{1, 3, 5\}$.

Is the outcome larger than 4? $C = \{5, 6\}$.

Is the outcome equal to 3? $D = \{3\}$.

Each question corresponds to a subset of the sample space Ω .

Such subsets are called *events*.

Why Set Operations?

Once questions are represented as sets, combinations of questions correspond naturally to set operations.

Let

$\{2, 4, 6\}$

$A = \{\text{outcome is even}\},$

$\{4, 5, 6\}$

$B = \{\text{outcome is larger than 3}\}.$

Then

$$A \cap B = \{\text{outcome is even AND larger than 3}\} = \{4, 6\},$$

$$A \cup B = \{\text{outcome is even OR larger than 3}\} = \{2, 4, 5, 6\},$$

and

$$A^c = \{\text{outcome is NOT even}\} = \{1, 3, 5\}.$$

Thus,

AND $\longleftrightarrow \cap$

OR $\longleftrightarrow \cup$

NOT $\longleftrightarrow ^c$

From Questions to Probabilities

Probability assigns a number to each event. For example,

$$A = \{\text{outcome is even}\} = \{2, 4, 6\},$$

and for a fair die,

$$P(A) = \frac{3}{6} = \frac{1}{2}.$$

Therefore,

Question	→	Event (set)	→	Probability.
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This is why set theory provides the mathematical language underlying probability theory.

Before introducing probability spaces, we briefly review several fundamental concepts.

Definition 1. *A set is a collection of distinct objects, called elements.*

Sets can be specified in two common ways.

Method 1: Listing Elements

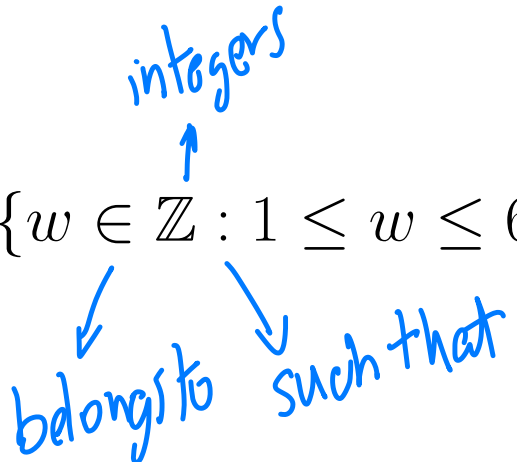
For example,

$$\Omega = \{1, 2, 3, 4, 5, 6\},$$

which represents the sample space of rolling a fair die.

Method 2: Set Builder Notation

The same set can be written as

$$\Omega = \{w \in \mathbb{Z} : 1 \leq w \leq 6\},$$


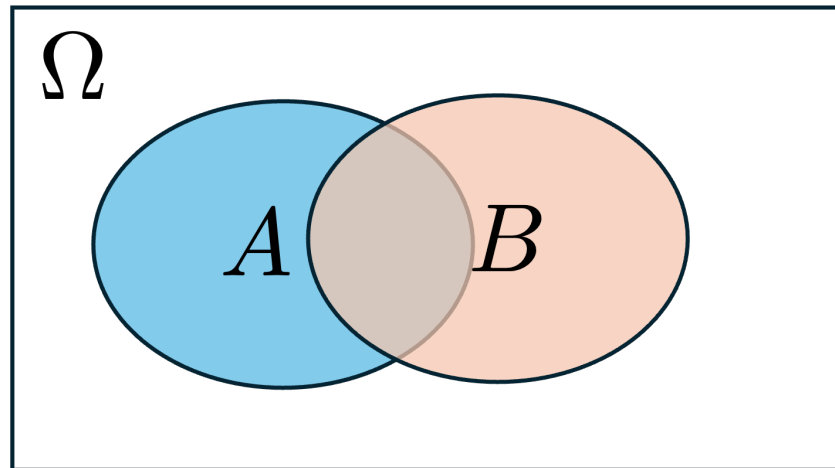
where

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

denotes the set of integers.

Venn Diagrams

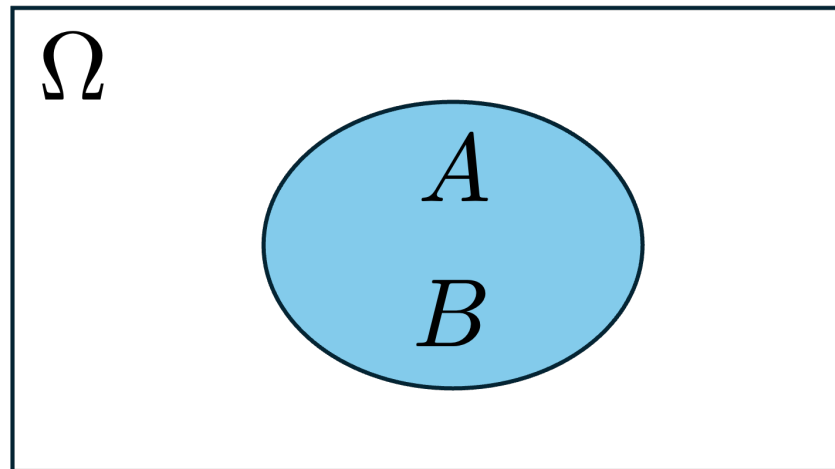
Venn diagrams provide graphical illustrations of relationships among sets.



Equality and Inclusion of Sets

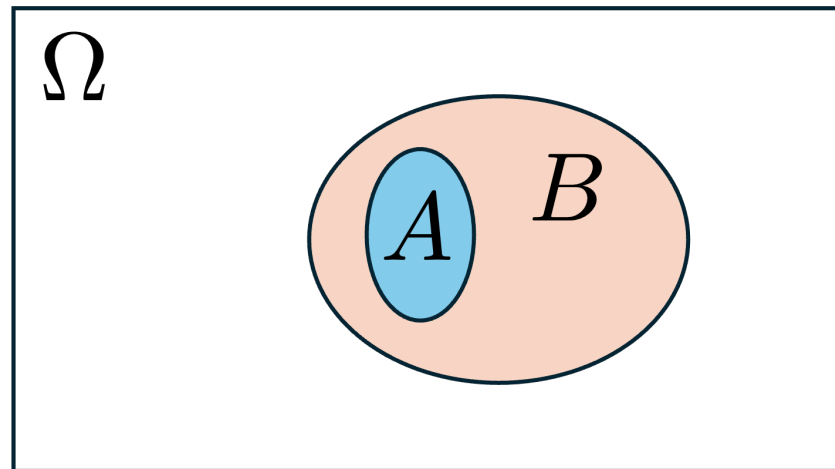
Definition 2. *Two sets A and B are equal ($A = B$) if they contain exactly the same elements. Equivalently,*

$$\forall w : w \in A \iff w \in B.$$



Definition 3. A set A is called a subset of B ($A \subseteq B$) if every element of A is also an element of B . Equivalently,

$$\forall w : w \in A \implies w \in B.$$



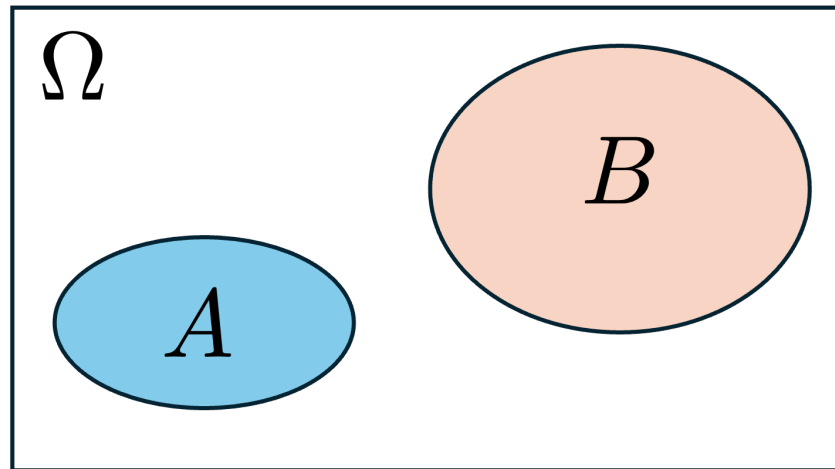
Definition 4. *The empty set is denoted by*

$$\emptyset = \{\}.$$

It contains no elements.

Definition 5 (Disjoint Sets). *Two sets are called disjoint if*

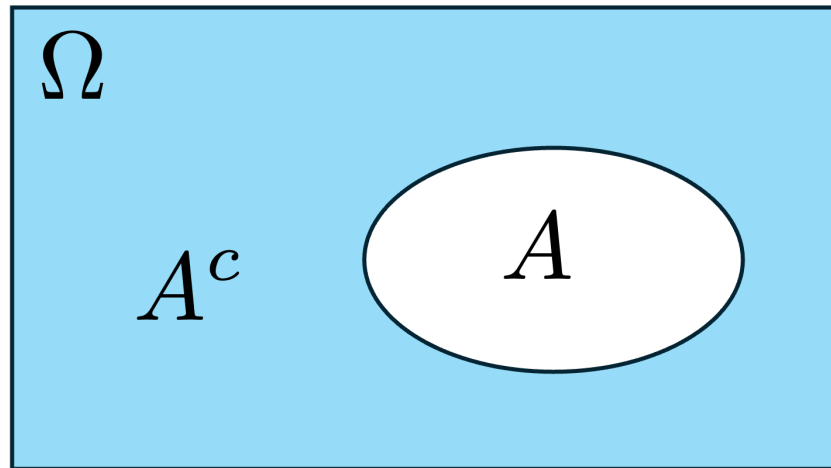
$$\forall w : w \in A \implies w \notin B.$$



Set Operations

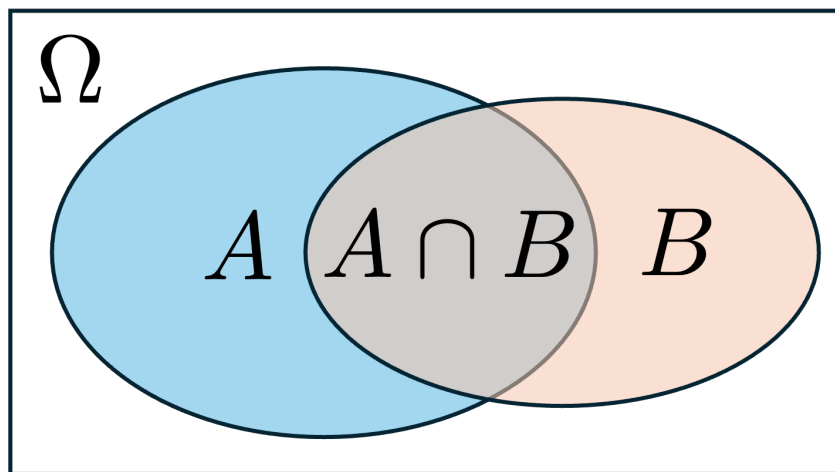
Definition 6 (Set complement). *The complement of a set is*

$$A^c = \{\omega \in \Omega : \omega \notin A\}.$$



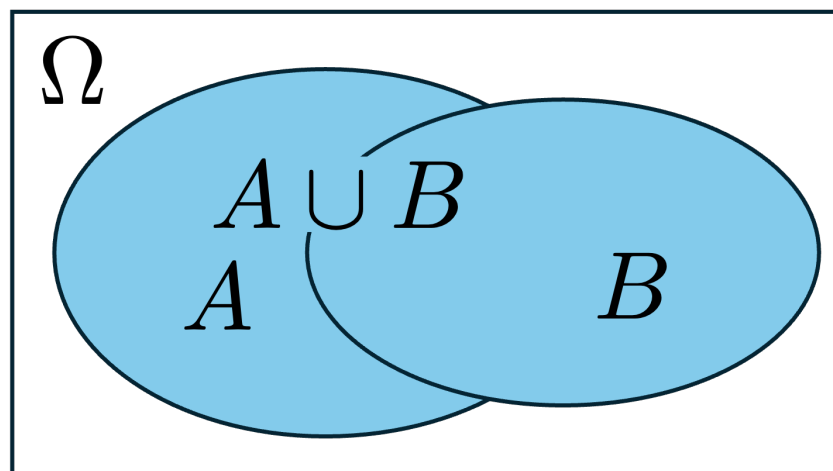
Definition 7 (Intersection). *The intersection of two sets is*

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}.$$



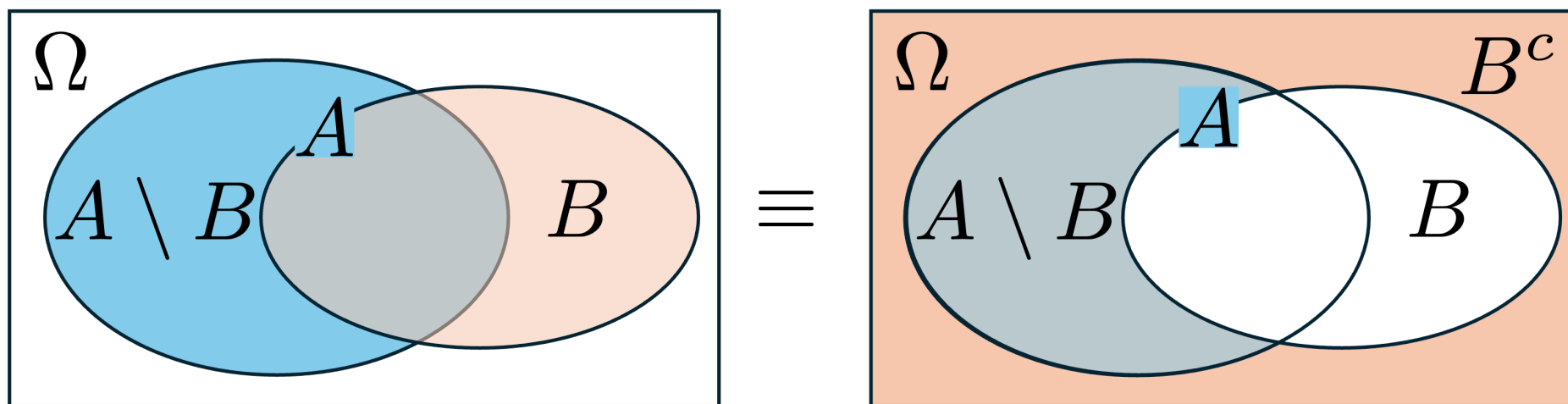
Definition 8 (Union). *The union of two sets is*

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}.$$



Definition 9 (Set Difference). *The difference between two sets is*

$$A \setminus B = \{\omega \in \Omega : \omega \in A, \omega \notin B\} = A \cap B^c.$$



Algebra of Sets

The following identities will be used throughout the course.

Commutative Laws

$$A \cup B = B \cup A,$$

(+)

$$A \cap B = B \cap A.$$

(x)

Associative Laws

$$A \cup (B \cup C) = (A \cup B) \cup C, \quad 1 + (2 + 3) = (1 + 2) + 3$$

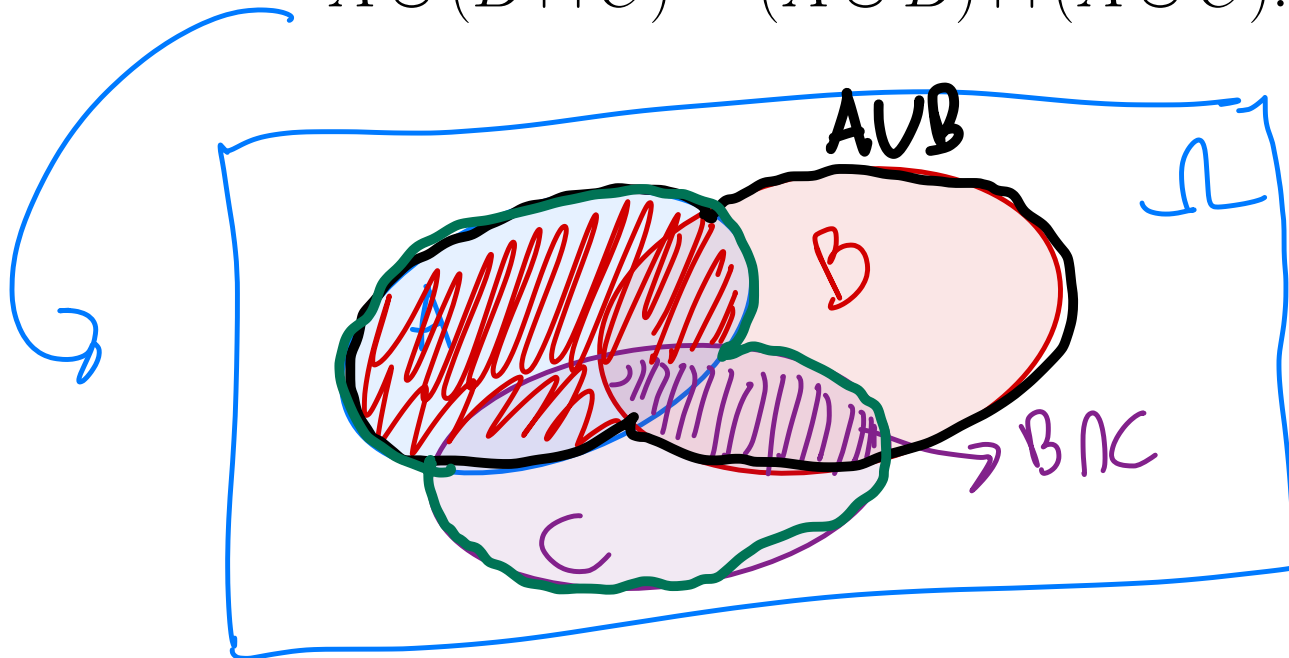
$$A \cap (B \cap C) = (A \cap B) \cap C. \quad 1 \times (2 \times 3) = (1 \times 2) \times 3$$

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$2 \times (3 + 4) \\ = (2 \times 3) + (2 \times 4)$$

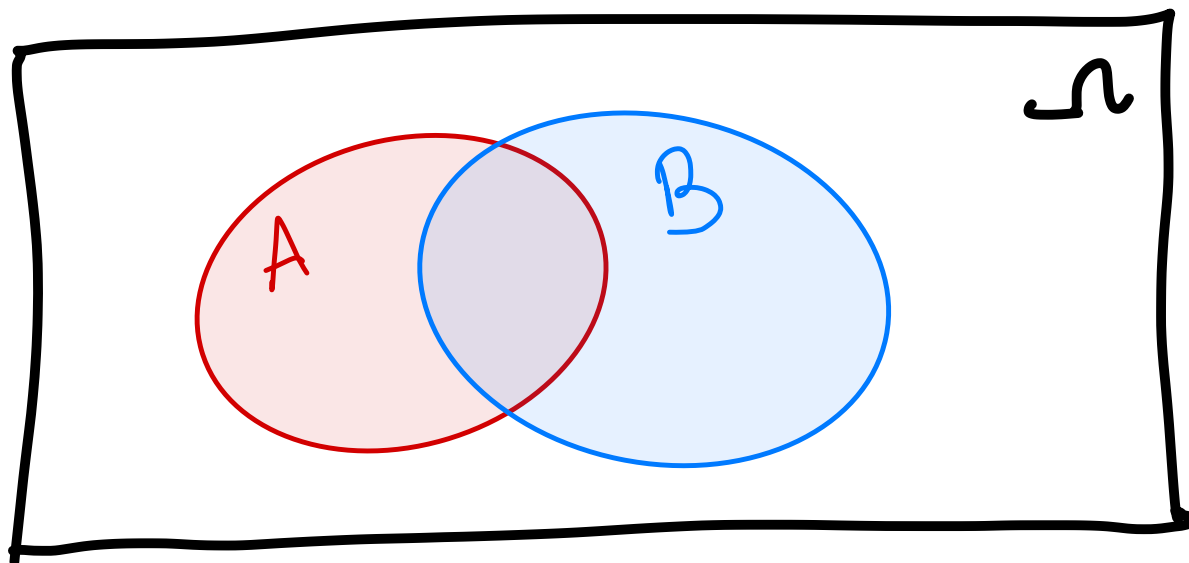
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$



De Morgan's Laws

$$(A \cap B)^c = A^c \cup B^c,$$

$$(A \cup B)^c = A^c \cap B^c.$$



Complement Laws

$$(A^c)^c = A,$$

$$A \cup A^c = \Omega,$$

$$A \cap A^c = \emptyset.$$

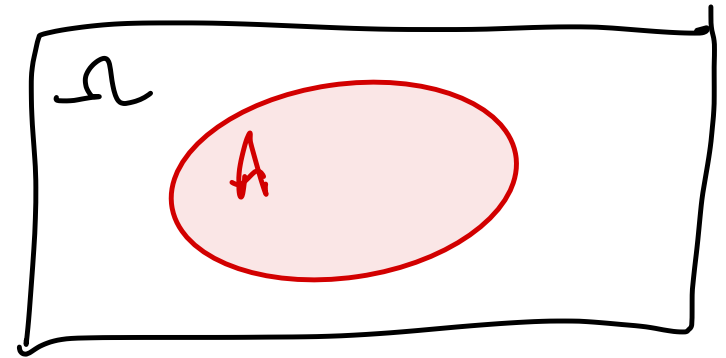
Identity Laws

$$A \cap \Omega = A,$$

$$A \cup \Omega = \Omega,$$

$$A \cap \emptyset = \emptyset,$$

$$A \cup \emptyset = A.$$



Finite, Countable, and Uncountable Sets

Definition 10. *A set is finite if it contains a finite number of elements.*

Example: $\Omega = \{1, 2, 3, 4, 5, 6\}$.

Definition 11. *A set is countably infinite if its elements can be placed into one-to-one correspondence with the positive integers.*

Examples include

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Z}_+$, $\mathbb{Q}$ (rationals)

Definition 12. *A set that is not countable is called uncountable.*

Examples include

$$\mathbb{R}, \quad [a, b], \quad (a, b).$$

$$\begin{array}{c} [0, 1] \\ \downarrow \\ 0 \leq x \leq 1 \end{array}$$

$$\begin{array}{c} (0, 1) \\ \downarrow \\ 0 < x < 1 \end{array}$$

$$\begin{array}{c} [0, 1) \\ \downarrow \\ 0 \leq x < 1 \end{array}$$

Observe that

Every uncountable set is infinite.

However,

Not every infinite set is uncountable.

For example, $\mathbb{N}$ is infinite but countable.

This distinction will later separate discrete and continuous probability models.

Power set

Definition 13 (Power Set). *The power set of Ω , denoted by 2^Ω , is the set of all possible subsets of Ω :*

$$2^\Omega = \{A : A \subseteq \Omega\}.$$

Essentially, it encodes all possible questions about the outcome.

Example:

$$\Omega = \{1, 2\},$$

$$2^\Omega = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Index Sets

Definition 14. *Let I be a set. A family indexed by I is a collection*

$$\{A_i : i \in I\},$$

where each $i \in I$ labels one member A_i . The set I is called the index set.

In essence, an index set specifies how a collection is indexed.

Examples:

$$\{x[n] : n \in \mathbb{Z}\},$$

$$\{x(t) : t \in \mathbb{R}\}.$$

Frequently used index sets include

$$\mathbb{N} = \{1, 2, 3, \dots\},$$

$$\mathbb{N}_0 = \{0, 1, 2, \dots\},$$

$$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\},$$

$$\mathbb{Z}_+ = \{0, 1, 2, \dots\},$$

$$\mathbb{R} = (-\infty, \infty),$$

$$\mathbb{R}^+ = [0, \infty).$$

Indexed Families of Sets

Let

$$\{A_i : i \in I\}$$

be an indexed family of sets.

The union is defined by

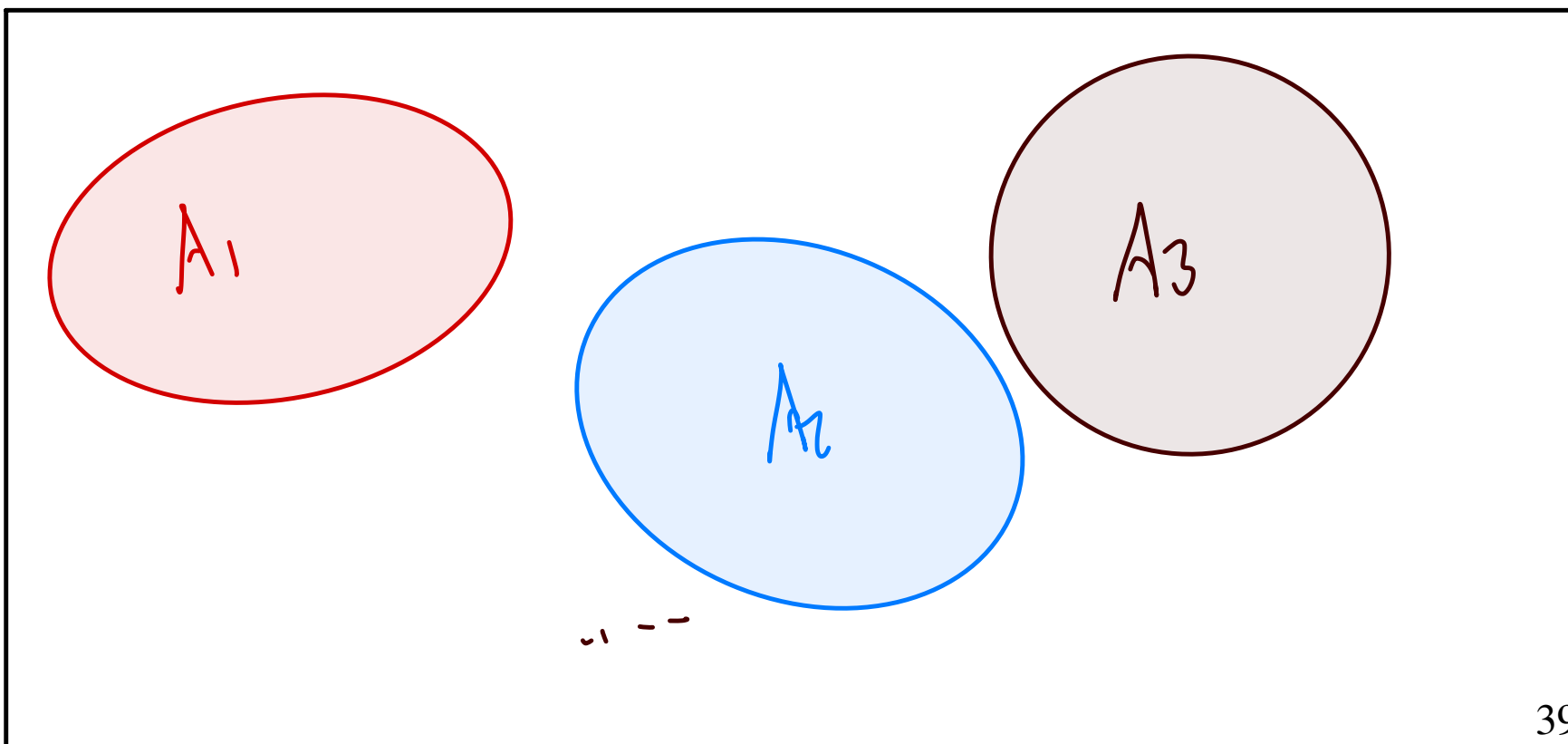
$$\bigcup_{i \in I} A_i = \{w : w \in A_i \text{ for at least one } i \in I\}.$$

The intersection is defined by

$$\bigcap_{i \in I} A_i = \{w : w \in A_i \text{ for every } i \in I\}.$$

An indexed family is called *disjoint* if

$$A_i \cap A_j = \emptyset, \quad i \neq j.$$



Partition

A collection $\{A_i : i \in I\}$ is called a *partition* of Ω if

$$A_i \cap A_j = \emptyset, \quad i \neq j, \quad \text{and} \quad \bigcup_{i \in I} A_i = \Omega.$$

